

Grade 7 Accelerated
Unit 1
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

Jaxon solved the inequality $32 < -4(-1.4n - 22)$, but made an error. His work is shown.

- What error did Jaxon make?
He divided twice on the right side of the inequality. He should have only divided once causing the -4 to cancel out therefore not affecting the terms within the parenthesis.
- Write a brief note to Jaxon explaining how to avoid making the same mistake in the future. Be sure to include what he did correctly in your note.

Jaxon's work	
$32 < -4(-1.4n - 22)$	
$\frac{32}{-4} < \frac{-4}{-4} \left(\frac{-1.4n}{-4} - \frac{22}{-4} \right)$	
$-8 < 1(0.35n + 5.5)$	
$-8 < 0.35n + 5.5$	
$-5.5 \quad -5.5$	
$-13.5 < 0.35n$	
$0.35 \quad 0.35$	
$-\frac{270}{7} < n$	

Answers may vary. Jaxon, the left side of your inequality was worked out correctly. Next time, be sure to divide only the term being distributed by itself as well as on the other side of the inequality to maintain equal actions on both sides.

- Solve Jaxon's inequality correctly and represent the solution on a number line.

$$\frac{32}{-4} < \frac{-4(-1.4n - 22)}{-4} \quad \frac{14}{-1.4} < \frac{-1.4n}{-1.4}$$

$$-8 < -1.4n - 22$$

$$+22 \quad +22 \quad -10 < n$$

For questions 4-7, solve each inequality and represent the solution on a number line.

- $$\frac{3}{5} \cdot \frac{5}{3} \left(-a + \frac{3}{4} \right) \leq -2 \cdot \frac{3}{5}$$

$$-a + \frac{3}{4} \leq \frac{-6}{5} - \frac{3}{4}$$

$$\frac{-3}{4} \quad -1 \cdot -a \leq \frac{-39}{20} \cdot -1$$

$$-a \leq \frac{-24}{20} - \frac{15}{20} \quad a \geq 1 \frac{19}{20}$$

- $$-3 \left(2t - \frac{4}{3} \right) > 6$$

$$2t - \frac{4}{3} < -2 + \frac{4}{3}$$

$$+ \frac{4}{3} \quad 2t < \frac{-6}{3} + \frac{4}{3}$$

$$\frac{2t}{2} < \frac{-2}{3} \div 2 \quad t < -\frac{1}{3}$$

- $$\frac{-6.2+m}{4} > -3.3$$

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$$4 \left(\frac{-6.2+m}{4} \right) > -3.3 \cdot 4$$

$$-6.2 + m > -13.2$$

$$+6.2 \quad +6.2 \quad m > -7$$

- $$\frac{2}{5}k - 8.4 \leq -4$$

$$+8.4 \quad +8.4$$

$$5 \cdot \frac{2}{5}k \leq 4.4 \cdot 5$$

$$2k \leq 22$$

$$\frac{2k}{2} \leq \frac{22}{2} \quad k \leq 11$$

Use the following scenario for questions 8–10.

When a store had sold $\frac{3}{5}$ of the shirts that were on display, they brought out another 40 from the stock room. The store likes to keep at least 170 shirts on display. The manager wrote the inequality $\frac{2}{5}x + 40 \geq 170$, where x represents the number of shirts.

8. Explain what $\frac{2}{5}$ means in the inequality.

$\frac{2}{5}$ represents the amount of shirts left from the original whole after $\frac{3}{5}$ were sold.

9. Solve the inequality.

$$\frac{2}{5}x + 40 \geq 170$$

$$\begin{array}{r} -40 \quad -40 \\ \hline \end{array}$$

$$5 \cdot \frac{2}{5}x \geq 130 \cdot 5$$

$$\frac{2x}{2} \geq \frac{650}{2}$$

$$x \geq 325$$

10. Explain what the solution means in terms of the situation.

The solution represents how many shirts the store started with.